

Mastering Quantum Mechanics — Problem 9.1

Harmonic oscillators beyond the turning points

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Problem 9.1

Harmonic oscillators beyond the turning points.

1. Consider the simple harmonic oscillator energy eigenstate with $n = 0$. Calculate the probability $P(\text{out} | n = 0)$ that the distance $|x|$ to the origin takes a value greater than the amplitude of a *classical* oscillator of the same energy (the particle is thus “out” of the classically allowed region). Write the answer in terms of an integral over a dummy variable u ; the integral defines a function related to the error function $\text{erf}(x)$. Evaluate it numerically.
2. Repeat for the energy eigenstates $n = 1$ and $n = 2$; find $P(\text{out} | n = 1)$ and $P(\text{out} | n = 2)$.

Solution

Setup and notation

We use Zwiebach’s conventions for the harmonic oscillator. The energy spectrum is (Eq. (9.2.30))

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \dots \quad (1)$$

The normalized energy eigenstates are (Eqs. (9.2.35)–(9.2.36))

$$\varphi_n(x) = N_n H_n(x/L_0) e^{-x^2/2L_0^2}, \quad L_0 = \sqrt{\frac{\hbar}{m\omega}} \quad (\text{Eq. (9.1.14)}), \quad (2)$$

with H_n the Hermite polynomials (Eq. (9.2.33)): $H_0 = 1$, $H_1 = 2u$, $H_2 = 4u^2 - 2$. It is cleanest to work in the dimensionless coordinate

$$u \equiv \frac{x}{L_0}, \quad dx = L_0 du. \quad (3)$$

Using the Hermite orthogonality relation $\int_{-\infty}^{\infty} H_n(u)^2 e^{-u^2} du = 2^n n! \sqrt{\pi}$, the normalization $\int |\varphi_n|^2 dx = 1$ (Eq. (9.2.37)) fixes

$$N_n^2 L_0 = \frac{1}{2^n n! \sqrt{\pi}}. \quad (4)$$

Classical turning point of the same energy

A classical oscillator of energy E_n has amplitude A_n set by $\frac{1}{2}m\omega^2 A_n^2 = E_n$, so

$$A_n^2 = \frac{2E_n}{m\omega^2} = \frac{(2n+1)\hbar}{m\omega} = (2n+1) L_0^2 \quad \Longrightarrow \quad \boxed{u_n \equiv \frac{A_n}{L_0} = \sqrt{2n+1}}. \quad (5)$$

For $n = 0$ this is just $u_0 = 1$: the classical amplitude equals the oscillator size L_0 .

General probability outside the classical region

Since $|\varphi_n|^2$ is even, the “out” probability is twice the tail:

$$P(\text{out} | n) = 2 \int_{A_n}^{\infty} |\varphi_n(x)|^2 dx = \frac{2}{2^n n! \sqrt{\pi}} \int_{\sqrt{2n+1}}^{\infty} H_n(u)^2 e^{-u^2} du. \quad (6)$$

This is the requested integral form. We now evaluate it for $n = 0, 1, 2$ using the elementary tail integrals (repeated integration by parts)

$$\int_a^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2} \operatorname{erfc}(a), \quad (7)$$

$$\int_a^{\infty} u^2 e^{-u^2} du = \frac{a}{2} e^{-a^2} + \frac{\sqrt{\pi}}{4} \operatorname{erfc}(a), \quad (8)$$

$$\int_a^{\infty} u^4 e^{-u^2} du = \left(\frac{a^3}{2} + \frac{3a}{4} \right) e^{-a^2} + \frac{3\sqrt{\pi}}{8} \operatorname{erfc}(a). \quad (9)$$

$n = 0$ ($H_0 = 1$, $u_0 = 1$).

$$P(\text{out} | 0) = \frac{2}{\sqrt{\pi}} \int_1^{\infty} e^{-u^2} du = \operatorname{erfc}(1) = 1 - \operatorname{erf}(1) \approx \boxed{0.1573}. \quad (10)$$

$n = 1$ ($H_1^2 = 4u^2$, $2^1 1! = 2$, $u_1 = \sqrt{3}$).

$$P(\text{out} | 1) = \frac{4}{\sqrt{\pi}} \int_{\sqrt{3}}^{\infty} u^2 e^{-u^2} du = \operatorname{erfc}(\sqrt{3}) + \frac{2\sqrt{3}}{\sqrt{\pi}} e^{-3} \approx \boxed{0.1117}. \quad (11)$$

$n = 2$ ($H_2^2 = 16u^4 - 16u^2 + 4$, $2^2 2! = 8$, $u_2 = \sqrt{5}$). Combining the three tail integrals, the e^{-a^2} pieces collect to $(8a^3 + 4a)e^{-a^2}$ and the erfc pieces to $4\sqrt{\pi}$, so

$$P(\text{out} | 2) = \frac{1}{4\sqrt{\pi}} \left[(8a^3 + 4a)e^{-a^2} + 4\sqrt{\pi} \operatorname{erfc}(a) \right]_{a=\sqrt{5}} = \operatorname{erfc}(\sqrt{5}) + \frac{11\sqrt{5}}{\sqrt{\pi}} e^{-5} \approx \boxed{0.0951}. \quad (12)$$

Checks

- **Dimensions/units.** Each P is dimensionless (probabilities); all x -dependence was absorbed into the dimensionless $u = x/L_0$.
- **Monotonic decrease.** $P(\text{out} | n)$ falls with n : $0.157 \rightarrow 0.112 \rightarrow 0.095$. This is the correspondence principle at work—higher n is “more classical,” so less probability leaks past the turning point; as $n \rightarrow \infty$, $P(\text{out} | n) \rightarrow 0$.
- **Limits.** For $n = 0$ the answer reduces to a pure Gaussian tail $\operatorname{erfc}(1)$, consistent with φ_0 being a pure Gaussian (Eq. (9.2.40)).

$$\boxed{P(\text{out} | 0) \approx 0.157, \quad P(\text{out} | 1) \approx 0.112, \quad P(\text{out} | 2) \approx 0.095}$$